Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN_Shift-1_Batch-II** **Date: 20-01-2026**

Time: 09.00Am to 12.00Pm **GTM-16** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature: _____**

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20-01-2026_ Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main_Shift-1_Batch-II_GTM-16_Test Syllabus

MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. For the given sets

$A=\{x: x\in Z, x^2-5x+6=0\}$

$B=\{x^2 / x\in Z-\{0\}\}$

$C=\{x: x\in Z, x^2=36\}$

$D=\{x: x\in Z, x^2<25\}$

The infinite set is:

1) A **2)** B **3)** C **4)** D

2. Let R be a relation on N defined by $R=\{(a,b): a \text{ is a factor of } b\}$. Then

STATEMENT-1: R is reflexive and transitive.

STATEMENT-2: R is transitive and symmetric

1) Both statements are true

2) Both statements are false

3) Statement-1 is true and Statement-2 is false

4) Statement-1 is false, but statement-2 is true

3. $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ =$

1) $\frac{3}{4}$ **2)** $\frac{3}{8}$ **3)** $\frac{3}{16}$ **4)** $\frac{3}{32}$

4. $\sin\left(\frac{\pi}{4}\cot\theta\right)=\cos\left(\frac{\pi}{4}\tan\theta\right)\Rightarrow\theta=$

1) $n\pi+\frac{\pi}{3}, n\in Z$ **2)** $n\pi+\frac{\pi}{4}, n\in Z$ **3)** $n\pi+\frac{\pi}{6}, n\in Z$ **4)** $n\pi+\frac{\pi}{2}, n\in Z$

5. If $\frac{\pi}{2}<\alpha<\frac{3\pi}{2}$, then the modulus – amplitude form of $(1+\cos 2\alpha)+i\sin 2\alpha$ is

1) $-2\cos\alpha\left[\cos(\pi+\alpha)+i\sin(\pi+\alpha)\right]$

2) $2\cos\alpha\left[\cos\alpha+i\sin\alpha\right]$

3) $2\cos\alpha\left[\cos(-\alpha)+i\sin(-\alpha)\right]$

4) $-2\cos\alpha\left[\cos(\pi-\alpha)+i\sin(\pi-\alpha)\right]$

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6. sum of the non-real roots of $(x^2+x-2)(x^2+x-3)=12$ is

1) -1 **2)** 1 **3)** -6 **4)** 6

7. A bag contains n white and n black balls. Pairs of balls are drawn without replacement until the bag is empty. If the number of ways in which each pair consists of one black and one white ball is 576, then n=

1) 4 **2)** 8 **3)** 16 **4)**24

8. If the number of five-digit numbers with distinct digits and 2 at the 10th place is 336k, then k is equal to

1) 8 **2)** 7 **3)** 4 **4)** 6

9. If the sum of the coefficients in the expansion of $(k^2x^2-6kx+5)^{37}$ vanishes, then k =

1) – 1 or -5 **2)** -2 or -3 **3)** 1 or 5 **4)** 2 or 3

10. If $x=\sum_{n=0}^{\infty}a^n, y=\sum_{n=0}^{\infty}b^n, z=\sum_{n=0}^{\infty}c^n$, where a,b,c are in A.P such that $|a|<1, |b|<1$ *and* $|c|<1$ then x,y,z are in

1) A.P **2)** G.P **3)** H.P **4)** A.G.P

11. If the points (1,2) and (3,4) were to be on the same side of the line $3x-5y+a=0$, then

1) 7 < a < 11 **2)** a = 7 **3)** a = 11 **4)** a < 7 or a > 11

12. If $x^2+y^2+2gx-8y+4g^2-7g+18=0$ represents a circle then g∈

1) [1/3, 2] **2)** R – [1/3, 2] **3)** R **4)** (0,2)

13. The equation of the parabola is given by $y^2+8x-12y+20=0$. Tick the correct option given below.

1) Vertex (-2,6) **2)** Focus (0, 6) **3)** Latus rectum 4 **4)** Axis x=2

14. The eccentric angle of a point on the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ at a distance $\frac{5}{4}$ units from the focus on the positive X-axis is

1) $\frac{\pi}{3}$ **2)** $\frac{\pi}{4}$ **3)** $\frac{\pi}{6}$ **4)** $\cos^{-1}\left(\frac{3}{4}\right)$

15. $\lim_{x\to -1^+}\frac{\sqrt{\pi}-\sqrt{\cos^{-1}x}}{\sqrt{1+x}}=$

1) $\frac{\sqrt{2}}{\sqrt{\pi}}$ **2)** $\frac{1}{\sqrt{2\pi}}$ **3)** $\frac{\sqrt{\pi}}{\sqrt{2}}$ **4)** $\sqrt{2\pi}$

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16. If the mean deviation about the mean of n consecutive integers is $\frac{8(n+1)}{n}$, then n =

1) 27 **2)** 30 **3)** 33 **4)** 36

17. If two points are P (7,-5,11) and Q (-2,8,13), then projection of PQ on a straight line with direction cosines $\frac{1}{3},\frac{2}{3},\frac{2}{3}$ is

1) 7 **2)** 8 **3)** 9 **4)** None of these

18. If $x^y = e^{x-y}, then\ \frac{dy}{dx} =$

1) $(1+\log x)^{-1}$ **2)** $(1+\log x)^{-2}$ **3)** $\log x(1+\log x)^{-2}$ **4)** $\log x(1+\log x)^{-1}$

19. The value of $\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ is

1) $\frac{26}{25}$ **2)** $\frac{25}{26}$ **3)** $\frac{50}{51}$ **4)** $\frac{52}{51}$

20. Let $f(\alpha)=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$, then $(f(\alpha))^{-1}$ is equal to:

1) $f(\alpha)$ **2)** $f(-\alpha)$ **3)** $f(\alpha-1)$ **4)** $f(\alpha+1)$

SECTION-II

(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. The maximum value of the expression $\frac{1}{\sin^2\theta+3\sin\theta\cos\theta+5\cos^2\theta}$ is

22. $\int\sqrt{1+\sec x}\,dx$ is equal to $\frac{k}{101}\tan^{-1}\sqrt{\frac{1-\cos x}{\cos x}}+C$, then the value of k is

23. Suppose for every integer n, $\int_n^{n+1} f(x)dx = n^2$. The value of $\int_{-2}^{4} f(x)dx$

24. If $y(x)$ is solution of $(x+1)\frac{dy}{dx}-xy=1, y(0)=-1,$ then the value of $y\left(\frac{-6}{5}\right)$ is equal to

25. A four digit number is formed by the digit 1,2,3,4,5,6 and 8 (repetition allowed). The probability that the number has odd digits at both ends is $\frac{p}{q}$ (in lowest from, p,q$\in N$), then $q-p$ is

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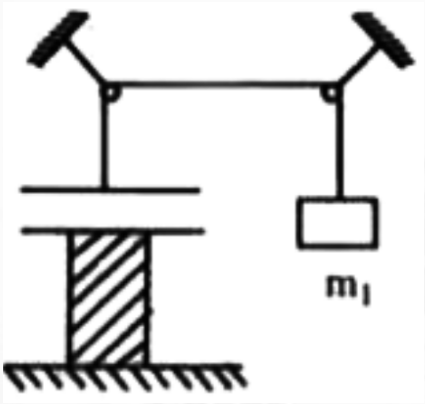
PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

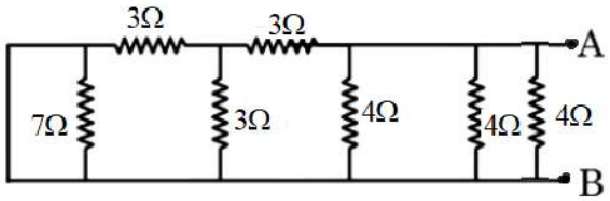
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. In the given figure a capacitor of plate area A is charged upto charge q. the mass of each plate is m_2. The lower plate is rigidly fixed, the value of m_1 if the system remains in equilibrium is



1) $m_2+\frac{q^2}{\epsilon_0 Ag}$ **2)** m_2 **3)** $\frac{q^2}{2\epsilon_0 Ag}+m_2$ **4)** $2m_2$

27. Equivalent resistance of the following network is



1) 28Ω **2)** 4Ω **3)** 7Ω **4)** $\frac{36}{35}\Omega$

28. A solid conducting sphere and a hallow conducting spherical shell of equal diameters are raised to same surface potential from zero potential then charges given to them

1) Both have equal charge **2)** Only hollow sphere has charged

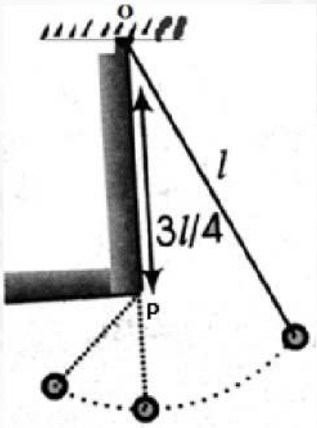
3) Solid sphere has more charge **4)** Hollow sphere has more charge

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JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
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29. A series combination of coil of inductance L and series resistance 12Ω is connected across a AC source RMS 12V. 50 Hz supply. Calculate approximate value of L, if circuit RMS current is 0.5 A

1) 0.66 H **2)** 0.0066 H **3)** 0.066 H **4)** 6.6 H

30. A simple pendulum has time period T for small oscillations. An obstacle p is situated below the point of suspension O at a distance $\frac{3l}{4}$. The pendulum is released from rest through out the motion the moving string makes small angle with vertical . find time after which the pendulum returns back to its initial position



1) $\frac{2T}{3}$ **2)** $\frac{4T}{3}$ **3)** $\frac{3T}{4}$ **4)** $\frac{5T}{4}$

31. Statement-I : The images formed due to total internal reflections are much brighter than those formed by mirrors or lenses

Statement-II: There is no loss of intensity in total internal reflection

1) It Both statements are TRUE and Statement-2 is the correct explanation of Statement-1

2) If both statements are TRUE and Statement-2 is not the correct explanation of STATEMENT-1

3) If statement-1 is true and Statement-2 is false

4) If Statement-1 is false, but statement-2 is true

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JEE MAIN 2023 RANK 1
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32. Assertion : Photo electric effect can only be explained by the wave nature of light.

Reason : For every metal there exists a limiting frequency of incident light called threshold frequency. Below which electron emission is not possible

1) A and R are true and R is correct explanation of A

2) A and R are true and R is not correct explanation of A

3) A is true, R is false

4) A is false, R is true

33. In YDSE, the two slits are separated by 0.1 mm and they are 0.5 m from the screen, the wavelength of light used is 5000 $\overset{0}{A}$. Find the distance between 7th maxima and 11th minima on the screen

1) 7.25 mm **2)** 8.75 mm **3)** 6.75 mm **4)** 5.25 mm

34. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The density of the copper $\left({}^{64}_{29}Cu\right)$ nucleus is greater than that of the carbon C nucleus.

Reason (R): The nucleus of mass number A has a radius proportional to $A^{1/3}$.

In the light of the above statements, choose the **most appropriate answer** from the options given below:

1)A is correct bus R is not correct

2)Both A and R are correct R is not the correct explanation of A

3)A is not correct but R is correct

4)Both A and R are correct and R is the correct explanation of A

35. Assertion: The angle of contact of mercury with glass is obtuse,

Reason: The cohensive forces between mercury molecules are stronger than the abhesive force between mercury and glass

1) A is true and R is false

2) A is false, but R is true

3) Both A and R are true and R is the correct explanation of A

4) Both A and R are true but R is not the correct explanation of A

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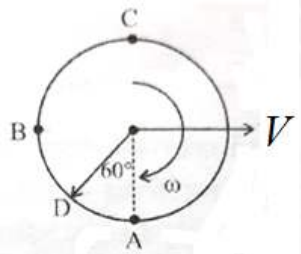
36. A satellite is to revolve around the earth is a circular orbit at a height of 800 km above the surface of the earth now what will be the time period (approximately) period? Take g at the surface $g = 9.8 m/\sec^2$ and radius of the earth 6400 km

1) 118 min **2)** 132 min **3)** 100 min **4)** 141 min

37. A plane flying horizontally with a speed of 216 km/h at an altitude of 500m, drops a package at an instant. The package hits the ground at a point P, 10s after it is dropped. Displacement of 'P' from the position of the plane where the package was released is; **(Use acceleration due to gravity g=10** m/s^2 and neglect air resistance)

1) $10\sqrt{61}m$ **2)** $100\sqrt{61}m$ **3)** $\sqrt{610}\,m$ **4)** $1000\sqrt{61}\,m$

38. A rigid body is rolling without slipping on the horizontal fixed surface, then match the column-I with Column-II and select the correct answer from the codes given below.



Column-I	**Column-II**
A. Velocity at point A, i.e v_A	1. $v\sqrt{2}$
B. Velocity at point B, i.e v_B	2. Zero
C. Velocity at point C, i.e v_c	3. v
D. Velocity at point D i.e v_D	4. 2v

1) **A**-2 ; B-1 ; C-4 ; D-3 **2)** A-1 ; B-3; C- 4; D- 2

3) A-4; B-3; C-2; D-1 **4)** A-2; B-3; C- 4; D-1

39. Assertion: When a solid sphere is placed in the fluid under high pressure, then it is compressed uniformly on all sides

Reason: The force applied by fluids acts in perpendicular direction at each point of surface that's why solid sphere is compressed uniformly

1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion

2) Both Assertion and Reason are correct but reason is not the correct explanation of Assertion

3) Assertion is correct but Reason is incorrect

4) Assertion is incorrect but Reason is correct

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40. In elastic collision,

Statement-I. In elastic collision,before collision intial kinetic energy of system is equal to final kinetic energy of the system after collision and total momentum is conserved.

Statement-II. In elastic collision, Kinetic energy during the collision time Δt is constant.

1) Statement I is ture, Statement-II is true

2) Statement-I is false, Statement-II is true

3) Statement-I True, Statement-II is false

4) Both statement I and II are False

41. A ring is cut form a platinum tube of 8.5 cm internal and 8.7 cm external diameter. It is supported horizontally from a pan of a balance so that it comes in contact with the water in a glass vessel. Calculate the surface tension of water if an extra 3.97 gmwt is required to pull it away from water. Take g=980 cm/s^2.

1) 72 $dynecm^{-1}$ **2)** 98 $dynecm^{-1}$ **3)** 83 $dynecm^{-1}$ **4)** 70 $dynecm^{-1}$

42. Consider sound waves travelling in ideal gases of Neon(Ne), Chlorine (Cl_2), and Methane (CH_4). All the gases have the same ratio $\frac{P}{\rho}$, Where P is the pressure and ρ is the density. The ratio of the speed of sound through the gases $V_{Ne} : V_{Cl_2} : V_{CH_4}$ is given by:

1) $\sqrt{\frac{5}{3}} : \sqrt{\frac{4}{3}} : \sqrt{\frac{7}{5}}$ **2)** $\sqrt{\frac{5}{3}} : \sqrt{\frac{7}{5}} : \sqrt{\frac{4}{3}}$ **3)** $\sqrt{\frac{7}{5}} : \sqrt{\frac{5}{3}} : \sqrt{\frac{4}{3}}$ **4)** $\sqrt{\frac{4}{3}} : \sqrt{\frac{7}{5}} : \sqrt{\frac{5}{3}}$

43. Find out the increase in moment of inertia I of a uniform rod of length l mass M (coefficient of linear expansion α about its perpendicular bisector when its temperature is slightly increased by ΔT

1) **2IαΔT**. **2)** $3I\alpha\Delta T$ **3)** **5IαΔT** **4)** . **IαΔT**

44. A steam pipe of radius 5 cm carries steam at 100^{0}C.The pipe is covered by a jacket of insulating material 2 cm thick having a thermal conductivity 0.07 $Wm^{-1}K^{-1}$.If the temperature at the outer wall of the pipe jacket is 20^{0}C how much heat is lost through the jacket per metre length in an hour? Take $\ln 1.4 = 0.336$

1) $4.76\times10^5 J$ **2)** $3.76\times10^5 J$ **3)** $5.76\times10^5 J$ **4)** $6.76\times10^5 J$

45. Assertion: The strain produced by a hydraulic pressure is volumetric in nature.

Reason: It is a ratio of change in volume ΔV to the original volume V.

1)Both Assertion and Reason are correct and Reason is the correct explanation of Assertion

2)Both Assertion and Reason are correct but Reason is not the correct explanation of assertion

3)Assertion is correct but Reason is incorrect

4)Assertion is incorrect but reason is correct

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SECTION-II
(NUMERICAL VALUE TYPE

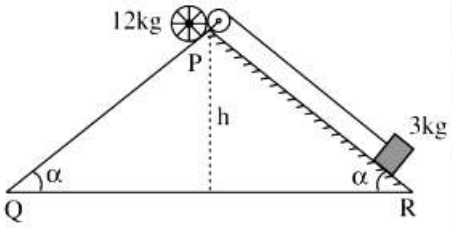
This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. An electron is released from rest near an infinite non-conducting sheet of uniform charge density '$-\sigma$' magnitude of the rate of change of de-Broglie wave length associated with the electron varies inversely as n^{th} h power of time at "t"sec. The numerical value of $|n|$ is _____.

47. A one meter long (both ends open) organ pipe is kept in a diatomic gas that has double the density of air at S.T.P. assuming the speed of sound in air at STP is 300 m/sec, the frequency difference between the fundamental and second harmonic of this pipe is _____ Hz (neglect the end correction at open end)

48. A rolling wheel of 12kg is on an inclined plane at position P and its centre connected to a mass of 3 kg through a string of fixed length passing over the fixed massless pulley as shown in figure. Consider PR as friction free surface.



The velocity of centre of mass of the wheel when it reaches at the bottom Q of the inclined plane PQ will be $\frac{1}{2}\sqrt{xgh}$ m/sec. the value of x is _____.

49. A solid sphere of radius R acquires a terminal velocity v_1 when falling (due to gravity) through a viscous fluid having a coefficient of viscosity η. The sphere is broken into 27 identical spheres. If each of these acquires a terminal velocity v_2, when falling through the same fluid, the ratio (v_1/v_2) equals

50. Suppose that the electric filed amplitude of an electromagnetic wave in vacuum is $E_0 = 120N/c$ and that its frequency is $f = 50.0M\ Hz$ determine B_0 value is _____ $\times 10^{-7} tesla$

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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. A 100 mL solution of 0.1 N HCl was titrated with 0.2 N NaOH solution. The titration was discontinued after adding 30 mL of NaOH solution. The remaining titration was completed by adding 0.25 N KOH solution. The volume of KOH required for completing the titration is

1) 70 mL **2)** 32 mL **3)** 35 mL **4)** 16 mL

52. 4d, 5p, 5f and 6p-orbitals are arranged in the order of decreasing energy. The correct option is

1) 6p > 5f > 5p > 4d **2)** 5p > 5f > 4d > 5p

3) 5f > 6p > 4d > 5p **4)** 5f > 6p > 5p > 4d

53. the correct order of decreasing electronegativity values among the elements A-beryllium, B-oxygen, C-nitrogen and D-magnesium is

1) B > C > A > D **2)** C > D > B > A **3)** A > B > C > D **4)** A > B > D > C

54. Which of the following pair of molecules contain odd electron molecule and an expanded octet molecule?

1) BCl_3 and SF_6 **2)** NO and H_2SO_4 **3)** SF_6 and H_2SO_4 **4)** BCl_3 and NO

55. A process will be spontaneous at all temperature if

1) $\Delta H>0$ and $\Delta S< 0$ **2)** $\Delta H<0$ and $\Delta S>0$

3) $\Delta H<0$ and $\Delta S<0$ **4)** $\Delta H>0$ and $\Delta S>0$

56. Match the following species with the corresponding conjugate acid.

	Species		Conjugate acid
A.	NH_3	1.	CO_3^{2-}
B.	HCO_3^-	2.	NH_4^+
C.	H_2O	3.	H_3O^+
D.	HSO_4^-	4.	H_2SO_4
		5.	H_2CO_3

CODES

1) A-4; B-3; C-2; D-1 **2)** A-2; B-5; C- 3; D-4

3) A-1; B-2; C-3; D- 4 **4)** A- 5; B-3; C-4; D-2

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57. Statement I: For $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ *if more* Cl_2 is added, the equilibrium will shift in backward direction hence, equilibrium constant will decrease.

Statement II Addition of inert gas to the equilibrium mixture at constant volume, does not alter the equilibrium.

1) Both Statement I and Statement II are false.

2) Statement I is true but Statement II is false.

3) Statement I is false but Statement II is true.

4) Both Statement I and Statement II are true.

58. Statement-I: HNO_2 acts both as an oxidising as well as a reducing agent.

Statement-II: The O.N of N can increase above +3 and can also decrease below +3.

1) Both Statement I and Statement II are false.

2) Statement I is true but Statement II is false.

3) Statement I is false but Statement II is true.

4) Both Statement I and Statement II are true.

59. Assertion(A)- Among isomeric pentanes,

2,2-dimethylpropane has highest boiling point.

Reason(R)- Branching decreases the boiling point.

1) Both A and R are true and R is the correct explanation of A.

2) Both A and R are true but R is not the correct explanation of A.

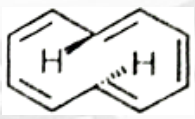
3) A is true but R is false.

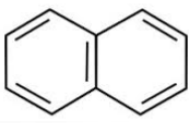
4) A is false but R is true.

60. Which of the following compounds is non-aromatic?

1) 

2) 

3) 

4) 

61. Given, $E^0_{(Cr^{3+}/Cr)} = -0.72V, E^0_{(Fe^{2+}/Fe)} = -0.42V$. The potential for the cell $Cr\left|Cr^{3+}(0.1M)\right|\left|Fe^{2+}(0.01\ M)\right|Fe$ is

1) – 0.339 V **2)** – 0.26 V **3)** 0.26 V **4)** 0.3 V

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62. For a chemical reaction, A and B. → Product the order is 1 with respect to both A and B

$Rate\ mol\ L^{-1}\ S^{-1}$	$[A]\ mol\ L^{-1}$	$[B]\ mol\ L^{-1}$
0.10	20	0.5
0.40	x	0.5
0.80	40	y

What is the value of x and y?

1) 80 and 4 **2)** 80 and 2 **3)** 40 and 4 **4)** 160 and 4

63. Assertion (A)- Cu cannot liberate hydrogen from acids.

Reason (R)- Cu has positive electrode potential.

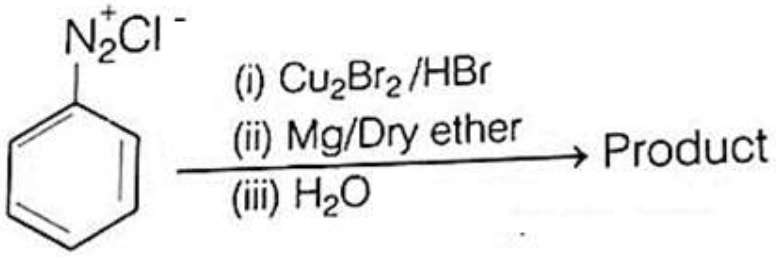
1) Both A and R are true and R is the correct explanation of A.

2) Both A and R are true but R is not the correct explanation of A.

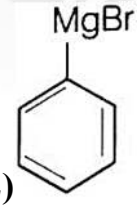
3) A is true but R is false.

4) A is false but R is true.

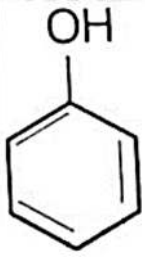
64. Identify the product in the following reaction.



1) 

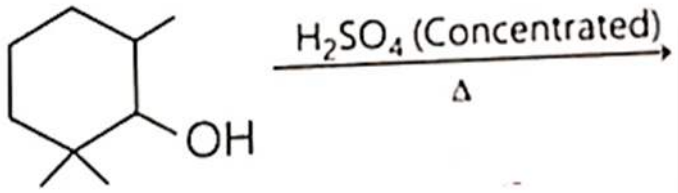
2) 

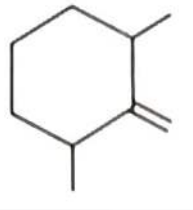
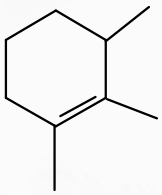
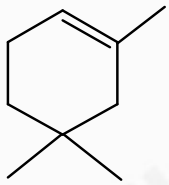
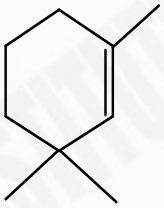
3) 

4) 

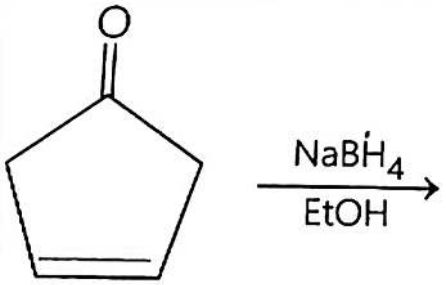
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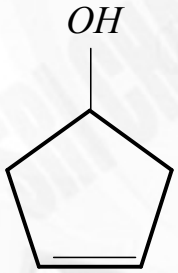
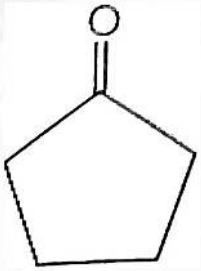
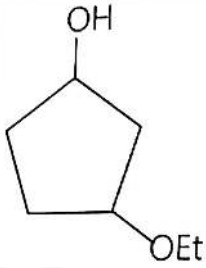
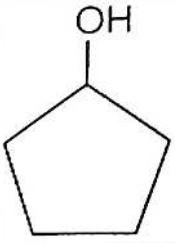
65. find out the major product from the following reaction



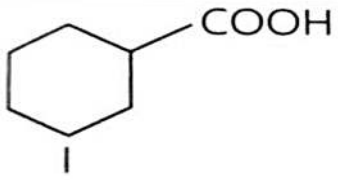
1)  **2)**  **3)**  **4)** 

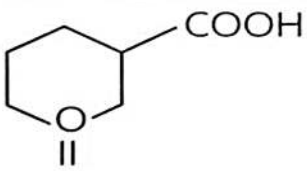
66. The major product of the following reaction is

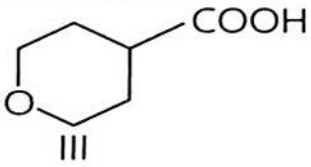


1)  **2)**  **3)**  **4)** 

67. The correct order of strengths of the carboxylic acids







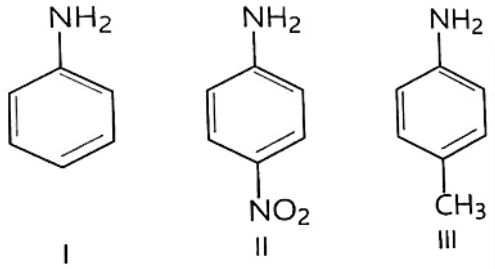
is

1) I > II > III **2)** II > III > I **3)** III > II > I **4)** II > I > III

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68. The correct increasing order of basic strength for the following compounds are



I II III

1) II < III < I **2)** III < I< II **3)** III < II < I **4)** II < I < III

69. Sulphur (S) containing amino acids from the following are

I. Isoleucine II. Cysteine III. Lysine IV. Methionine V. Glutamic acid

1) I, II, III **2) II, III, V** **3) II, IV** **4) I, IV**

70. Among halogens, the correct order of amount of energy released in electron gain enthalpy is

1) F > *Cl* > Br > I 2) F < *Cl* < Br < I 3) F < *Cl* > Br > I 4) F < *Cl* >Br < I

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

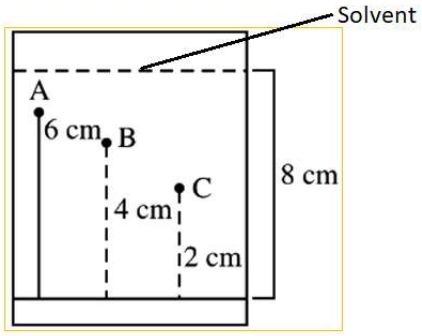
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. At 25°C and 1 atm pressure, the enthalpies of combustion are as given below.

Substance	H_2	C(graphite)	$C_2H_6(g)$
ΔH kJ mol-1	-286.0	-394.0	-1560.0

The enthalpy of formation of ethane is (-)_____KJ mol

72. Three organic compounds A, B and C were allowed to run in thin layer chromatography using hexane and gave the following result (see figure). The R_f value of the most polar compound is ..... x 10^{-2}.



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73. 150g of acetic acid was contaminated with 10.2g ascorbic acid $[C_6H_8O_6]$ to lower down its freezing point by $\left(X\times10^{-1}\right)^0 C$. The value of X is ……..(nearest integer)

$\left[given\ K_f=3.9K\ kg\ mol^{-1}, molar\ mass\ of\ ascorbic\ acid=176g\ mol^{-1}\right]$

74. half-life period of a first order reaction is 10 min. the percentage of the reaction will be completed in 100 min ……..(nearest integer)

75. $\left[Fe(CN)_6\right]^{4-}, \left[Ti(CN)_6\right]^{3-}, \left[Ni(CN)_4\right]^{2-}, \left[Fe(CN)_6\right]^{3-}, \left[Co(CN)_6\right]^{3-}$

Among the given complexes, number of diamagnetic complexes is ……

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